

**DEVELOPMENT OF A WEB-BASED UNIVERSITY ADMISSION
GUIDANCE AND PLACEMENT PLATFORM. A CASE STUDY OF
KAMPALA INTERNATIONAL UNIVERSITY, UGANDA.**

Submitted by:

MUGABI CALEB 2023-08-17673

ACAN LESLIE OBUA 2023-08-20750

MBABAZIZE JERRY DENZEL 2023-08-16896

**A PROJECT PROPOSAL SUBMITTED TO THE SCHOOL OF
MATHEMATICS AND COMPUTING IN PARTIAL FULFILLMENT
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OF BACHELOR OF INFORMATION TECHNOLOGY**

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DECLARATION

We, the undersigned, declare that this project proposal is our original work and has not been presented for any award of a degree in this or any other university. All sources of information used in this work have been properly acknowledged

Mugabi Caleb 2023-0817673

Signature:.....

Acan Leslie Obua 2023-08-20750

Signature:.....

Mbabazize Jerry Denzel 2023-08-16896

Signature:.....

APPROVAL

This project proposal titled “Development of a web-based university admission guidance and placement platform.” has been submitted with my approval as a supervisor.

Supervisor’s Name: Mr. ARIONG BEN

Signature:

Date:

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ABSTRACT

Access to suitable university program and career opportunities remains a major challenge for students in Uganda. Incoming students often struggle to select appropriate undergraduate program due to the complexity of Advanced Level subject combination eligibility rules set by the National Council for Higher Education (NCHE) and university-specific guidelines. This frequently results in mismatched choices, academic difficulties, delayed graduation and high dropout rates. Similarly, final year students face difficulties accessing centralized, relevant job and internship opportunities contributing to prolonged graduate unemployment, underutilized skills and limited career progression.

Current manual processes for course advising and career placement are inefficient, time consuming, inconsistent and error prone. Admissions staff spend excessive time on individual consultations while career services struggle to track applications, match skills to opportunities, and provide timely feedback. This creates communication gaps, frustration for students, and missed opportunities for both learners and the institution.

This project proposes the design and development of a web-based University Admission Guidance and Placement Platform to address these challenges. The system will enable incoming students to input their A-Level subject combinations (including principal subjects, General Paper, and subsidiary subjects on pass/fail basis) and grades to receive personalized program recommendations, complete with entry requirements, fees, duration and career prospects. Final year students will create profiles, upload CVs, browse posted opportunities, apply directly, track application status and receive notifications. Administrators will manage program data, opportunities and generate reports for decision making and institutional planning.

The study will adopt an incremental system development methodology incorporating requirements gathering through interviews, questionnaires and document review to ensure alignment with user needs and local practices. The system will be implemented using Python Flask for backend logic, MySQL for data management, Bootstrap for responsive frontend design and deployed on free cloud hosting. The expected outcome is a functional, user-friendly prototype that simplifies critical academic and career transitions, reduces manual workload, improves decision making and enhances graduate employability at KIU. The platform will contribute to better student support, institutional efficiency and alignment with national goals for quality higher education and skilled workforce development in Uganda.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The transition from secondary education to university and from university to employment remains one of the major challenges faced by students in Uganda. Globally, the process of higher education and career guidance has evolved significantly with the adoption of digital tools and platforms. In many developed countries, web-based systems allow students to explore program, check eligibility, compare options and connect directly to employers or internship providers. These platforms provide personalized recommendations, real-time updates and integrated tracking features that reduce uncertainty and improve outcomes.

In Uganda, however, the transition process is still largely manual and fragmented. Prospective university students often rely on word of mouth information from teachers, parents, peers or limited school guidance to understand program eligibility based on Advanced Level subject combinations. The National Council for Higher Education (NCHE) sets strict subject combination rules and minimum grade requirements yet many students lack clear, accessible information on how these rules apply to specific program at different universities. This leads to confusion, mismatched program choices, academic struggles, delayed graduation and high dropout rates, particularly in private universities where resources for individual advising are limited.

Final year students face similar difficulties in accessing centralized, relevant job and internship opportunities. Many depend on personal networks, occasional job fairs, informal postings, or scattered announcements from career offices. This approach is time-consuming, unreliable, limited in reach, and often excludes students without strong connections. As a result, graduate unemployment remains high, with many graduates taking extended periods to secure employment aligned with their qualifications, leading to underutilized skills and economic strain. The current manual processes for course advising and career placement are inefficient, inconsistent and lack personalization. Admissions staff spend considerable time on individual consultations, while career services struggle to track applications, match students to opportunities and provide timely feedback. This creates a significant communication gap between students, university offices and potential employers.

Therefore, there is a clear need for a web-based University Admission Guidance and Placement Platform that enables incoming students to input their A-Level subject combinations and grades to receive accurate, personalized program recommendations, complete with entry requirements, fees, duration, and career prospects. The platform will also allow final year students to create profiles, upload CVs, browse posted opportunities, apply directly, track application status and receive notifications. Administrators will be able to manage program data, opportunities and generate reports for institutional planning and decision making. Such a system will bridge existing gaps, reduce manual workload, improve decision making, enhance overall student support and graduate employability.

1.2 Statement of the Problem

Despite the increasing demand for structured guidance in higher education transitions, current methods at Kampala International University remain inefficient and unreliable. Incoming students lack access to accurate, up to date information on program eligibility based on Advanced Level subject combinations, entry requirements, fees and career prospects. This results in poor decision making, frequent program mismatches, academic underperformance, delayed graduation and high dropout rates.

Final year students also lack centralized access to relevant job and internship opportunities, leading to delayed employment, underutilized skills and prolonged unemployment. Many rely on informal networks or scattered announcements which are limited in reach and often outdated. Existing manual systems are time consuming, inconsistent and error prone. Admissions and career offices struggle to manage high volumes of inquiries, track applications, match skills to opportunities and generate meaningful reports. This creates significant communication gaps between students, university offices and potential employers resulting in wasted time, increased costs, frustration for students and missed opportunities for the institution.

The absence of an integrated digital platform to address both pre-enrollment guidance and post-graduation placement support worsens these issues highlighting the urgent need for a unified web-based solution.

1.3 Objectives of the Study

1.3.1 General Objective

To design and develop a web based platform that assists incoming students with course assignment based on Advanced Level subject combinations and connects final year students to relevant job and internship opportunities.

1.3.2 Specific Objectives

- 1) To investigate current challenges in course selection for incoming students and placement for final year students.
- 2) To analyze existing admission guidelines and placement practices at KIU and other universities to identify best practices and gaps.
- 3) To design a web system with Advanced level subject based course recommendation and career opportunity linking features.

1.3.3 Research Questions

- 1) What challenges do incoming students face in course selection and final year students face in placement support at KIU?
- 2) What are the key admission guidelines, placement practices and identified gaps from existing systems at KIU and other universities?
- 3) How can a web system be designed to provide accurate Advanced Level based course recommendations and effective career linking features?

1.4 Scope of the Study

1.4.1 Geographical Scope

The study will be conducted at Kampala International University (KIU) main campus, Uganda focusing on students and staff within the university environment.

1.4.2 System Scope

The system will:

- i) Allow input of Advanced Level subject combinations, grades, General Paper and subsidiary subjects (pass/fail) to generate program recommendations.
- ii) Display program details including entry requirements, fees, duration and career prospects.
- iii) Enable final year student profile creation with degree, skills and CV upload.
- iv) Allow career office administrators to post job/internship opportunities and manage applications.
- v) Provide application tracking, basic search/filtering, notifications, and reporting features.

1.5 Significance of the Study

This study is significant to various stakeholders including students, KIU admissions and career offices, the university management, researchers, system developers, policymakers and the broader higher education sector.

1) To incoming students:

The platform will simplify program selection by providing clear, personalized recommendations based on Advanced Level subject combinations reducing the risk of wrong choices and academic struggles.

2) To final year students:

The system will connect them to relevant job and internship opportunities, saving time, reducing stress and improving career decision making and employability.

3) To KIU Admissions Office:

The system will reduce manual advising workload, provide real-time data on program demand, enable better resource planning and improve efficiency in handling student intake.

4) To KIU Career Services Office:

The platform will streamline opportunity posting, application tracking and student matching, generate actionable placement statistics and help build stronger industry partnerships.

5) To KIU management:

The study will enhance student satisfaction and retention rates, strengthen KIU's reputation as a student-centered institution, contribute to higher graduate employability metrics and provide data for accreditation, ranking and strategic planning.

6) To the higher education sector:

The platform will offer a scalable, reusable digital model that other Ugandan universities can adopt to address similar transition challenges and support national goals for quality education and skilled workforce development.

7) To policymakers and government agencies:

The study will support the adoption of digital solutions in higher education, align with national ICT and education policies and demonstrate how technology can improve access to information and promote digital inclusion.

8) To researchers and academicians:

The study will add to existing literature on web based student support systems particularly within a local Ugandan context providing a reference point for future research on digital platforms, admission guidance and career placement.

9) To system developers:

The findings and system design will offer insights into best practices for developing user-friendly, scalable web applications tailored to local higher education needs.

10) To the community and society:

By improving access to accurate program and opportunity information, the study will contribute to reduced stress, better academic and career outcomes and stronger alignment between education and labor market needs.

11) To parents and guardians:

The platform will provide transparency and reassurance, allowing them to better support their children's academic and career decisions with reliable, up to date information.

12) To lecturers and academic advisors:

The system will reduce repetitive advising tasks, free up time for more focused academic mentoring, and provide data on student preferences and trends.

13) To industry partners and employers:

The platform will improve visibility of skilled graduates, facilitate direct access to talent, and enable more targeted recruitment and internship opportunities.

14) To KIU alumni:

The system can later be extended to include alumni networking, mentorship, and job referrals, strengthening lifelong connections with the university.

15) To ICT policy implementation:

The study demonstrates practical application of national ICT policies in higher education, showing how digital tools can enhance service delivery and efficiency.

16) To digital literacy promotion:

By requiring students and staff to interact with an online system, the platform will promote digital skills and familiarity with technology among users.

17) To transparency and accountability:

The system will standardize program and opportunity information, reduce misinformation, and build trust between students and university offices.

18) To economic development:

Improved transitions from education to employment will contribute to a more skilled workforce, supporting national economic growth and reducing graduate unemployment.

19) To environmental sustainability:

By reducing physical visits, printed materials, and repeated inquiries, the platform will contribute to lower paper use and travel-related emissions.

20) To emergency and relocation needs:

The system can assist students who require urgent program changes or career support e.g., transfers, sudden job needs providing quick, centralized access to information.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to the proposed web based University Admission Guidance and Placement Platform. The purpose of the literature review is to examine what other researchers and developers have done in the area of web based student support systems, identify strengths and weaknesses in existing solutions and establish gaps that justify the need for the current study. The review focuses on studies related to course recommendation systems, career placement platforms, digital student transition support and communication between students and university offices.

2.2 Related Studies

Several studies have been conducted on web based systems for student admission guidance and career placement. Research indicates that online platforms improve accessibility to academic and career information by allowing students to explore program and opportunities remotely.

According to Laudon and Laudon (2018), web based information systems enhance efficiency by centralizing data and enabling faster communication between students and university offices. Other studies emphasize the importance of personalized recommendation features in academic systems. Systems that incorporate rule based or simple logic for program suggestions provide students with clearer understanding of eligibility and career prospects, which improves decision making and satisfaction. Sommerville (2016) notes that well designed interfaces and accurate information increase system usability and user trust.

A study by Nielsen (2020) on usability and user experience found that systems incorporating clear displays of requirements and prospects help users better understand options before making decisions. The study emphasizes that structured information improves user engagement and confidence. This relates to the current research as the proposed system will provide detailed program recommendations based on Advanced Level subject combinations including entry requirements and career prospects.

According to a study by the World Bank Group (2023) on digital platforms in developing countries, locally tailored information systems are more effective than generalized global platforms. The study highlighted that systems designed for local contexts improve adoption, usability, and service delivery. This finding directly relates to the current study, which focuses on developing a platform tailored to KIU and Ugandan A-Level/NCHE guidelines.

A study by Pressman and Maxim examined the role of software engineering practices in developing reliable information systems. The study emphasized that structured system development methodologies improve system quality, maintainability, and user satisfaction. These findings relate to the current study as the proposed platform will follow a structured development process to ensure reliability and ease of use (Pressman & Maxim, 2020).

Another study by Sommerville focused on the development of web based systems and highlighted the importance of requirement analysis and system testing in producing effective software solutions. The study concluded that well tested web applications reduce user errors and system failures. This supports the current study's emphasis on testing and validation of the admission and placement platform (Sommerville, 2020).

A usability study by Nielsen revealed that user-centered design significantly improves the effectiveness of web applications. The research showed that systems with simple navigation and clear interfaces enhance user satisfaction and task completion rates. This study relates to the current research since the proposed platform will be designed with a simple interface to support ease of use by incoming and final-year students (Nielsen, 2020).

Research by the Uganda Communications Commission showed that increased internet penetration in urban Uganda has created opportunities for web-based service platforms. The study noted that digital systems can effectively support service access in urban communities. These findings justify the feasibility of implementing an online admission and placement system at KIU (Uganda Communications Commission, 2023).

A study by UNESCO on ICT and sustainable development highlighted that digital platforms enhance transparency and efficiency in service delivery. The research emphasized the role of ICT in improving access to information and communication between stakeholders. This supports the relevance of developing a web-based platform for student transitions (UNESCO, 2023).

Another study by the OECD examined the impact of digital transformation on information access and management. The study found that digital systems reduce operational costs and improve service responsiveness. This relates to the current study as the proposed platform aims to streamline course advising and placement coordination processes (OECD, 2023).

A study by Kothari on research methodology emphasized the importance of appropriate data collection and analysis methods in system development studies. The research highlighted that well designed methodologies improve the validity and reliability of study outcomes. This supports the methodological approach adopted in the current research (Kothari, 2019).

A study by Hevner et al. on design science research in information systems found that problem driven system development leads to practical and applicable solutions. The study emphasized developing systems that directly address real world problems. This aligns with the current study, which seeks to solve student transition challenges through a practical digital platform (Hevner et al., 2004).

Research by Turban, Pollard, and Wood examined the role of information systems in service management and found that web-based platforms significantly improve service coordination and user interaction. The study highlighted that online systems enhance efficiency and reduce delays in service delivery. This relates to the current study as the proposed platform will improve coordination between students, admissions staff, and career office administrators (Turban et al., 2018).

A study by Shneiderman et al. investigated user interface design principles and their impact on system usability. The study concluded that well designed interfaces improve user engagement and task performance. This supports the design of the proposed platform, which emphasizes simplicity and clear navigation for users (Shneiderman et al., 2018).

A study by Krug focused on usability in web applications and revealed that users prefer systems that are simple and intuitive. The research emphasized minimizing user effort and reducing complexity in system design. This is relevant to the current study as the platform will be designed to be easy to use by students with varying levels of digital literacy (Krug, 2014).

A study by the Interaction Design Foundation explored human-computer interaction in web-based systems and found that user-centered design increases adoption and satisfaction. The study highlighted the importance of understanding user needs during system development. This relates to the current study as user needs will be considered in designing the admission and placement platform (Interaction Design Foundation, 2024).

Research by the United Nations on e-government initiatives showed that digital platforms enhance access to information and improve service delivery at the institutional level. The study emphasized that online systems reduce barriers associated with manual processes. This supports the relevance of implementing a web-based platform at KIU (United Nations, 2022).

Research by Twinomugisha and Mwesigwa (2023) examined digital tools for student support in Ugandan universities. The study found that online platforms can significantly reduce confusion in programme selection by providing structured eligibility information and personalized suggestions. The authors emphasized that systems tailored to local A-Level rules improve student satisfaction and reduce dropout rates. This directly relates to the proposed platform, which will use A-Level subject combinations to generate accurate programme recommendations at KIU.

Okello and Ochieng (2022) investigated graduate employability in Uganda and highlighted the role of university career services. Their findings showed that centralized digital platforms for job postings and application tracking increase placement success rates by 30–40% in institutions with strong online systems. The study noted gaps in private universities, where manual processes dominate, supporting the need for an integrated career placement module in the proposed system.

Ssekamatte (2021) analyzed A-Level to university transitions in Uganda and identified poor access to accurate eligibility information as a major barrier. The research recommended digital tools to bridge this gap, particularly for private institutions with diverse programme offerings. This aligns with the current study's focus on A-Level-based recommendation features to address transition challenges.

Muwonge and Ssenkusu (2024) conducted a case study on online career guidance platforms in East African universities (Makerere and Kyambogo). They found that systems with profile creation, CV upload, and real-time opportunity matching improve student–employer connections. The study concluded that simple, mobile-friendly interfaces are critical for adoption in resource-constrained settings, a principle the proposed platform will follow.

Adetimirin (2022) reviewed digital support systems in African universities post-COVID-19. The research showed that web-based platforms reduce administrative workload and improve student access to services. The study stressed the importance of user-centered design and local relevance, findings that inform the proposed system's development approach.

Chawinga and Chipeta (2023) explored digital inclusion in higher education in Malawi, with lessons applicable to Sub-Saharan Africa. They found that locally adapted systems enhance usability and service delivery compared to generic platforms. This supports the need for a KIU-specific platform aligned with Ugandan NCHE guidelines.

Oyelere et al. (2021) conducted a systematic review of mobile learning and digital support in developing countries. They concluded that web and mobile-integrated systems increase student engagement and access to guidance. The study highlighted the importance of simple interfaces and notifications, features the proposed platform will incorporate.

Pimmer and Pachler (2022) reviewed mobile learning in African higher education and found that digital platforms improve transitions by providing timely information and interaction. Their findings justify the inclusion of notifications and tracking in the proposed system.

Al-Emran and Shaalan (2021) reviewed the technology acceptance model in educational contexts. They found that perceived usefulness and ease of use are key drivers of adoption for student support systems. This relates to the proposed platform's focus on intuitive design and clear value for students and staff.

Kabanda and Chipangura (2021) conducted a systematic review of digital transformation in Sub-Saharan African higher education. They identified that localized, integrated systems for academic and career support lead to higher satisfaction and efficiency. This supports the proposed platform's dual focus on admission guidance and placement.

2.3 Research Gaps Identified

Despite the availability of various online student support and placement systems, several gaps remain. Most existing platforms focus on either admission guidance or career placement, but not both in an integrated way leaving students to navigate two separate processes without continuity. Few systems in Uganda incorporate Advanced Level subject combinations including principal subjects, General Paper, and subsidiary subjects on pass/fail basis as the primary criterion for program recommendation despite its critical role in NCHE and university eligibility rules. This leads to incomplete or inaccurate guidance for incoming students often resulting in mismatched choices or repeated applications.

Current solutions often lack real time application tracking, personalized matching based on student skills, degree, interests and comprehensive reporting features for university decision making. Many platforms are not tailored to private universities like KIU where student populations, admission processes and industry partnerships differ significantly from public institutions. There is limited evidence of piloted systems that include multimedia or detailed program information e.g., fees, duration, career prospects alongside career linking features. Additionally, existing systems frequently overlook accessibility for students with varying digital literacy levels, fail to provide notifications for deadlines or status updates and do not support basic analytics for institutional planning e.g., program demand trends, placement success rates, or dropout patterns linked to program mismatch. Poor integration with local contexts such as Ugandan NCHE guidelines, university-specific admission practices or regional labor market needs further limits their applicability and effectiveness. Many platforms also lack mobile responsiveness or offline capabilities which are essential in areas with inconsistent internet access.

These gaps collectively highlight the need for a more holistic, context-specific solution. There is a clear absence of platforms that combine accurate pre-enrollment guidance with seamless post-graduation placement in a single, user-friendly system designed for Ugandan universities. The lack of emphasis on usability testing with diverse student groups, long-term impact evaluation, and institutional data utilization further restricts the ability of existing systems to deliver measurable improvements in student outcomes and administrative efficiency.

These gaps justify the need for the design and development of a web-based platform that is simple, locally adaptable, fully integrated and capable of improving access to program information, career opportunities, communication efficiency and institutional data insights. The proposed system will address these deficiencies by offering personalized, NCHE-compliant recommendations, real-time tracking, notifications, reporting and a focus on ease of use for all stakeholders.

CHAPTER THREE: METHODOLOGY

Introduction

This chapter explains the methods that will be used to carry out the study. It describes the research approach, target population, sampling techniques, sample size, system development methodology, data collection methods, system design and interface techniques, testing procedures, tools for implementation, logistical and ethical considerations, and other research techniques. The methodology is designed to ensure the proposed web-based University Admission Guidance and Placement Platform is user-centered, feasible within the available time and resources, and capable of effectively addressing the identified challenges.

3.1 Research Approach

The study will adopt a **Design Science Research (DSR)** approach (Hevner et al., 2004). This methodology is chosen because the core contribution of the study is the creation and evaluation of an IT artifact the web-based platform itself to solve real world problems in student admission guidance and career placement. DSR is particularly suitable for this project as it combines rigorous scientific methods with practical problem solving. The researcher will identify the problem, design a solution, build the system and evaluate its effectiveness through user feedback and performance metrics. This approach aligns with the incremental nature of the development process and allows iterative refinement based on emerging findings.

3.2 Target Population

The target population will include:

- i) Incoming Advanced Level students and new university students (estimated 3,000–4,000 annually based on recent enrolment trends).
- ii) Final year undergraduate students across various faculties (estimated 2,000–3,000).
- iii) Staff members in the Admissions Office and Career Services Office (approximately 20–30 personnel).

These groups are directly affected by or involved in the processes of program selection, eligibility checking, profile management, opportunity posting, application handling and student transition support. The population will be drawn from KIU main campus, Uganda with particular focus on faculties that have high student intake and placement activity.

3.3 Sampling Techniques

The study will employ **purposive sampling** combined with **convenience sampling**. Purposive sampling will be used to deliberately select participants who have direct experience with the current challenges e.g., final year students who are actively seeking placement, incoming students confused about program eligibility and staff members currently handling admissions or career services. Convenience sampling will supplement this by including readily available and willing participants during on-campus sessions ensuring a practical balance between relevance and accessibility within the limited timeframe.

3.4 Sample Size

The sample size will target approximately 100–120 participants, broken down as follows:

- i) 40–50 incoming/new students for A-Level recommendation testing and feedback.
- ii) 40–50 final year students for profile creation, application testing and placement feedback.
- iii) 15–20 staff from Admissions and Career Services Offices for requirements gathering and usability evaluation.

This size is manageable within the project timeline, sufficient for meaningful pilot testing, and allows for qualitative depth through interviews and quantitative breadth through surveys/questionnaires.

3.5 System Development Methodology

The study will adopt the **Incremental Development Methodology** also known as incremental prototyping, an iterative approach that builds the system in small, functional increments (Pressman & Maxim, 2020; Sommerville, 2020). Each increment will deliver a working subset of the system, allowing early testing, user feedback and refinement before proceeding to the next increment.

This methodology is chosen for the following reasons:

- i) It supports rapid delivery of usable components e.g., first the authentication and recommendation module, then profile and placement, then reports and notifications).
- ii) It reduces risk by identifying and fixing issues early in small cycles rather than at the end.
- iii) It accommodates user feedback throughout development ensuring the final system meets actual needs.
- iv) It is feasible within the limited timeframe and aligns well with the prototyping nature of final year projects.

The incremental process will follow these main cycles:

- 1) Increment 1: Core authentication, A-Level input and basic program recommendation.
- 2) Increment 2: Student profile creation, CV upload, opportunity browsing and application.
- 3) Increment 3: Admin functions (program/opportunity management), notifications, and reporting.
- 4) Increment 4: Full integration, usability testing, bug fixing, and final deployment.

3.5.1 Data Collection Methods

Data will be collected using a combination of primary and secondary methods:

- i) **Interviews:** Semi-structured interviews will be conducted with 15–20 staff members to gain in-depth insight into current processes, pain points, and desired features.
- ii) **Questionnaires/Surveys:** Structured questionnaires will be administered to 80–100 students to quantify challenges, preferences, and expectations.
- iii) **Observation:** Direct observation of current advising and placement sessions to identify real-world inefficiencies.
- iv) **Document Review:** Review of NCHE guidelines, KIU prospectus, career services reports, and related academic literature to inform requirements and validate design decisions.

3.5.2 System Design and Interface Techniques

System design will involve:

- i) Creating entity-relationship diagrams for database structure.

- ii) Developing data flow diagrams and use case diagrams for system processes.
 - iii) Designing wireframes and mockups for key user interfaces (login, A-Level input form, recommendation results, profile page, opportunity listing, admin dashboard).
- Interface design will follow **user-centered design principles** (Nielsen, 2020; Krug, 2014), prioritizing simplicity, clarity, mobile responsiveness, and minimal cognitive load to accommodate students with varying digital literacy levels.

3.5.3 Testing

Testing will occur at multiple levels:

- 1) Unit Testing: Individual modules e.g., recommendation logic, CV upload will be tested for correctness.
- 2) Integration Testing: Combined modules will be tested to ensure seamless interaction (e.g., login → profile → apply).
- 3) Usability Testing: 20–30 students and 10 staff will participate in pilot sessions to assess ease of use, navigation, and satisfaction (using think-aloud protocol and Likert-scale feedback).
- 4) Functional Testing: All features will be verified against requirements.
- 5) Security & Performance Testing: Basic checks for input validation, session security, and response time under load.

3.5.4 Tools for Implementation

The system will be implemented using:

- i) Backend: Python 3.11 with Flask framework (lightweight and rapid development).
- ii) Database: MySQL (relational, scalable, and suitable for structured data).
- iii) Frontend: HTML5, CSS3, Bootstrap 5 (responsive, mobile-first design).
- iv) Development Environment: Visual Studio Code (with extensions for Python, Git, and debugging).
- v) Version Control: Git (GitHub for collaboration and backup).
- vi) Deployment: Free cloud hosting (Render or Heroku) for pilot demonstration.

3.6 Logistical and Ethical Considerations

Ethical Considerations Ethical considerations will include:

- i) Obtaining informed consent from all participants before interviews, surveys, or testing.
- ii) Ensuring confidentiality and anonymity of all user information e.g., anonymizing survey responses.
- iii) Using collected data strictly for academic purposes and destroying it after the study.
- iv) Complying with data protection policies and ethical guidelines for research involving human participants.

Logistical Considerations Logistical considerations will involve:

- i) Use of the researcher's personal PC for development and testing.
- ii) Internet access for research, documentation, online collaboration, and deployment.
- iii) Access to willing participants (students and staff) during on-campus sessions for interviews, surveys, and usability testing.
- iv) Free/open-source tools and cloud hosting to keep costs minimal.

3.7 Research Techniques

This study will employ several research techniques to collect, analyze and interpret data relevant to the development of the web-based University Admission Guidance and Placement Platform.

1) Interview Technique

Interviews will be conducted with selected staff from Admissions and Career Services Offices to obtain in-depth information about current course selection and placement practices and communication challenges. This technique will enable the researcher to gather detailed explanations and clarify responses that could not be adequately captured through questionnaires.

2) Observation Technique

Observation will be used to study existing course advising and placement processes. The researcher will observe how incoming and final year students currently navigate program selection and job opportunity access and how staff handle inquiries. This technique will help identify practical challenges and inefficiencies in the current system.

4) Document Review Technique

Relevant documents such as NCHE admission guidelines, KIU program handbooks, previous research reports, journals, textbooks and online resources will be reviewed to understand existing solutions and system development practices. This technique will provide a theoretical foundation for the study and guide system design decisions.

5) System Development Technique

The study will apply an incremental system development approach in which user requirements will be analyzed, the system will be designed in stages, implemented, tested, and evaluated iteratively. This technique will ensure that the developed platform addresses the identified problems and meets user needs effectively.

6) Survey Technique

The survey technique will be used to gather information from incoming and final year students regarding the challenges they face in course selection and placement support. Questionnaires will be designed with structured questions (Likert scale, yes/no, open-ended) to obtain consistent and relevant quantitative and qualitative data. This technique is chosen because it allows data to be collected from many respondents within a short period of time.

Conclusion

This methodology is feasible within the available time, resources, and expertise. By combining Design Science Research principles with incremental development, user-centered data collection and rigorous testing, the study will deliver a functional, practical, and effective platform that addresses real student and institutional needs.

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APPENDICES

Appendix A: Work Plan

The project will be completed in **7 weeks** as follows:

Week 1: Finalize proposal with supervisor feedback, project setup (Git repo, Flask skeleton, MySQL database connection), basic authentication (registration/login and roles).

Week 2: Requirements refinement (quick interviews/questionnaires with 10–15 students/staff), A-Level course recommendation module (input form, based matching logic and results display).

Week 3: Final year student profile creation (degree, skills, CV upload), career office opportunity posting module.

Week 4: Application submission and tracking, basic search/filter for program and jobs.

Week 5: Notifications (in-app or email), reporting (course recommendations and placement statistics), admin dashboard basics.

Week 6: Full testing (unit, integration, usability with 10–20 users), bug fixing, security checks, performance testing.

Week 7: Final polish (responsive UI, deployment to free host like Render/Heroku), documentation, presentation slides, final supervisor demo and submission.

Appendix B: Sample Questionnaire (for Students)

Section A: Demographic Information

- i) Year of study
- ii) Incoming/New
- iii) Final year
- iv) Faculty

Section B: Current Challenges (Likert scale: 1 = Strongly Disagree to 5 = Strongly Agree)

- i) I find it difficult to choose the right university program based on my A-Level subjects. (1–5)
- ii) I receive sufficient guidance on program eligibility and requirements. (1–5)
- iii) I have easy access to relevant job/internship opportunities through KIU. (1–5)

Section C: Expected Features

- i) How useful would automatic program recommendations based on A-Level subjects be? (1–5)
- ii) Would you use a platform to upload your CV and apply to jobs? (Yes/No/Maybe)

Appendix C: Placeholders

- 1) Login page with role-based redirection
- 2) A-Level input form (principal subjects, General Paper, subsidiary pass/fail)
- 3) Program recommendation results page
- 4) Final year student profile creation with CV upload
- 5) Job opportunity listing and apply button
- 6) Admin dashboard showing reports and opportunity management

